DG INTERNS HUB

Cybersecurity Internship

WEEK 9 ASSIGNMENT

SOC INTRODUCTION & SETUP

SNORT IDS MONITORING PROJECT

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Internship	Cybersecurity Internship
Week	Week 9
Assignment	SOC Introduction & Snort IDS Setup
Environment	VirtualBox, Ubuntu, Snort IDS
Report Date	05 November 2026

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1. Introduction to SOC

Task 1 | SOC fundamentals

What is a Security Operations Center?

A Security Operations Center (SOC) is a team and operational function responsible for continuously monitoring systems, networks, endpoints and security events. Analysts investigate alerts, validate incidents, coordinate containment and document findings.

Core SOC Activities

- Monitor security telemetry and network traffic.
- Triage and investigate suspicious alerts.
- Escalate confirmed incidents for containment and recovery.
- Improve detection rules and maintain investigation records.

Basic SOC Workflow

Collect > **Detect** > **Triage** > **Investigate** > **Respond**

In this assignment, Snort is used as a network intrusion detection system (IDS). It inspects observed packets against configured signatures and generates alerts when rules match.

2. SOC Analyst Roles

Task 1 | Analyst escalation model

L1 Analyst

Monitoring and initial triage

Reviews SIEM/IDS alerts, checks severity and basic indicators, eliminates obvious false positives and escalates suspicious activity.

L2 Analyst

Deep investigation

Correlates evidence, examines packet and host logs, determines impact and coordinates containment with other teams.

L3 Analyst

Advanced response and detection engineering

Handles complex investigations, threat hunting, malware analysis, tuning detection logic and improving SOC playbooks.

3. SIEM, IDS and IPS

Task 1 | Key security technologies

SIEM

Centralizes security logs, correlates events and supports investigation and reporting.

Example: Wazuh / Elastic-based monitoring

IDS

Observes activity and raises alerts when suspicious traffic or behavior matches a detection rule.

Example: Snort in passive IDS mode

IPS

Operates inline and can actively drop or block matching traffic according to policy.

Example: Snort in an appropriately configured inline mode

4. Virtual Lab Architecture

Task 2 | Network monitoring design



This is a conceptual workflow diagram. The Ubuntu VM and Snort installation are verified. Custom-rule testing has not yet been evidenced.

5. Ubuntu Environment Setup

Task 2 | Real environment evidence

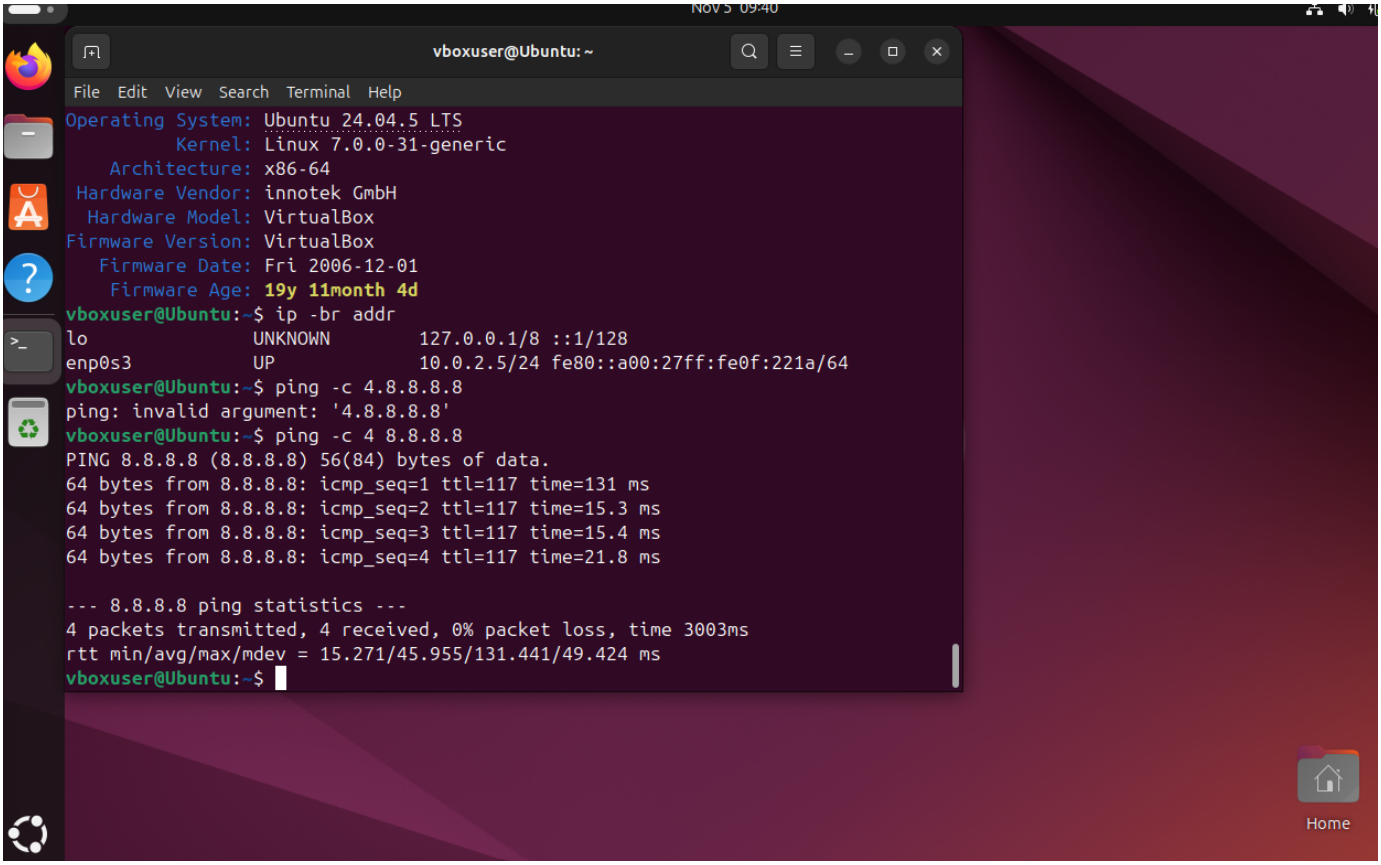


Figure 1. Ubuntu system details, network interface and successful ping test.

Observed environment: Ubuntu 24.04.5 LTS, VirtualBox, interface enp0s3, IP address 10.0.2.5/24. A ping to 8.8.8.8 returned 4 replies with 0% packet loss.

6. Snort Installation

Task 3 | Real package installation evidence

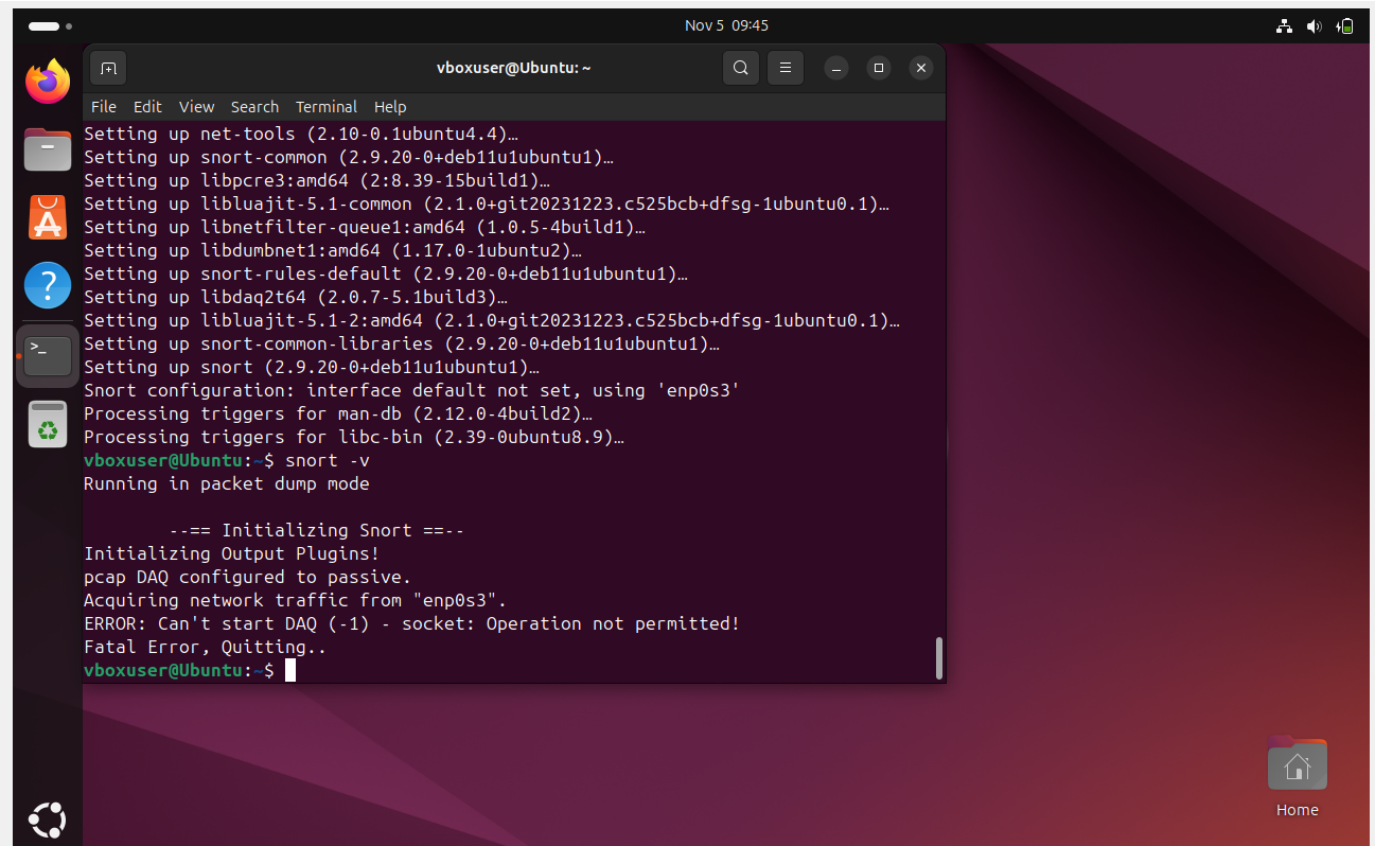


Figure 2. Snort package installation completed; enp0s3 selected.

Installation and Verification

```
sudo apt update
sudo apt install snort -y
snort -V
```

The installed Snort version is 2.9.20 GRE (Build 82). The earlier lowercase -v invocation attempted packet capture and failed without sufficient permissions; uppercase -V correctly reports version information.

7. Network Configuration

Task 2 / Task 3 | Verified HOME_NET value

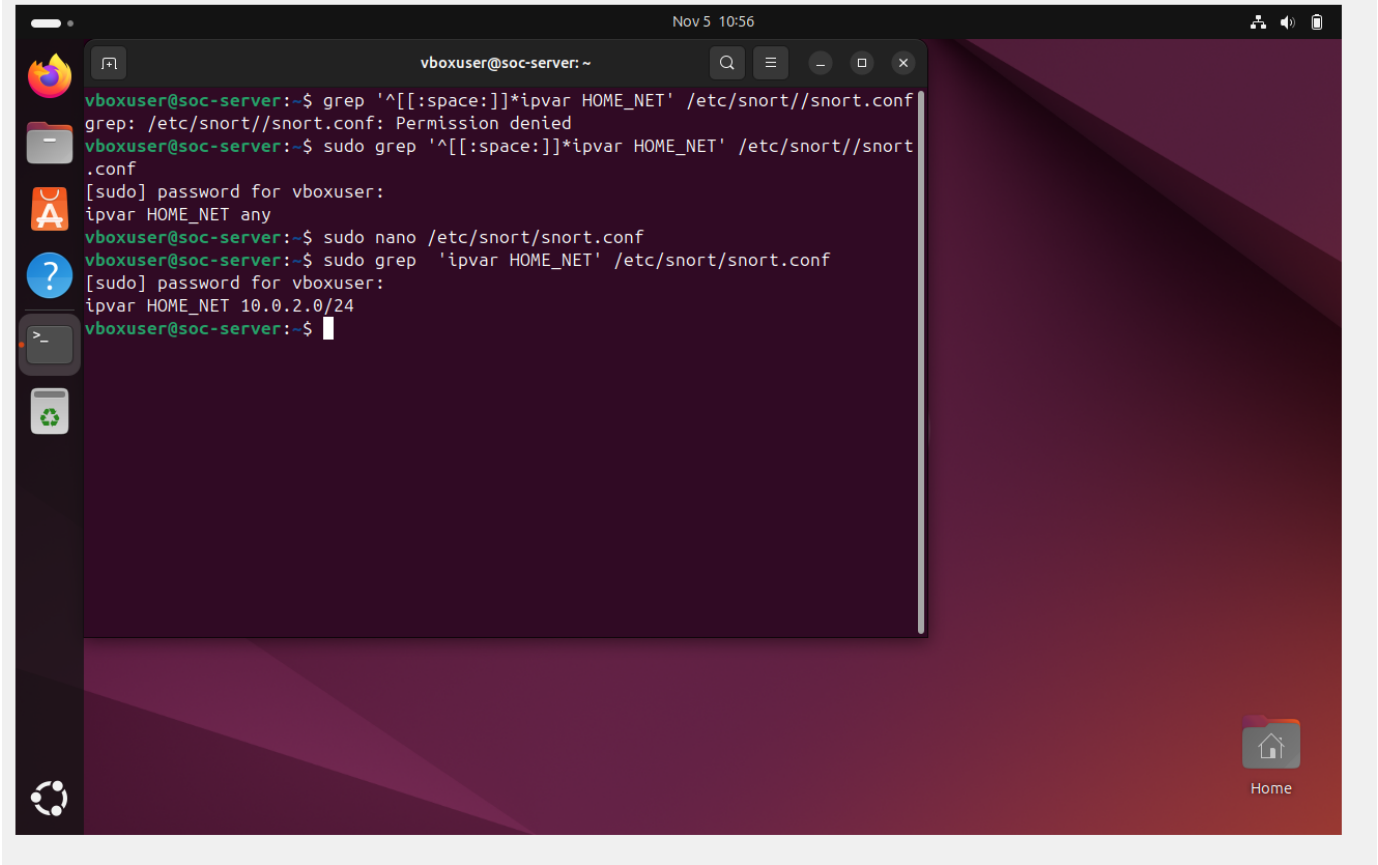


Figure 3. HOME_NET changed and verified in snort.conf.

Configuration Values

```
ipvar HOME_NET 10.0.2.0/24
ipvar EXTERNAL_NET any
```

The configured HOME_NET corresponds to the previously observed VM subnet. The interface name is enp0s3. Network ranges may need adjustment if the VirtualBox adapter settings change.

8. Snort Configuration Test

Task 3 | Verified successful validation

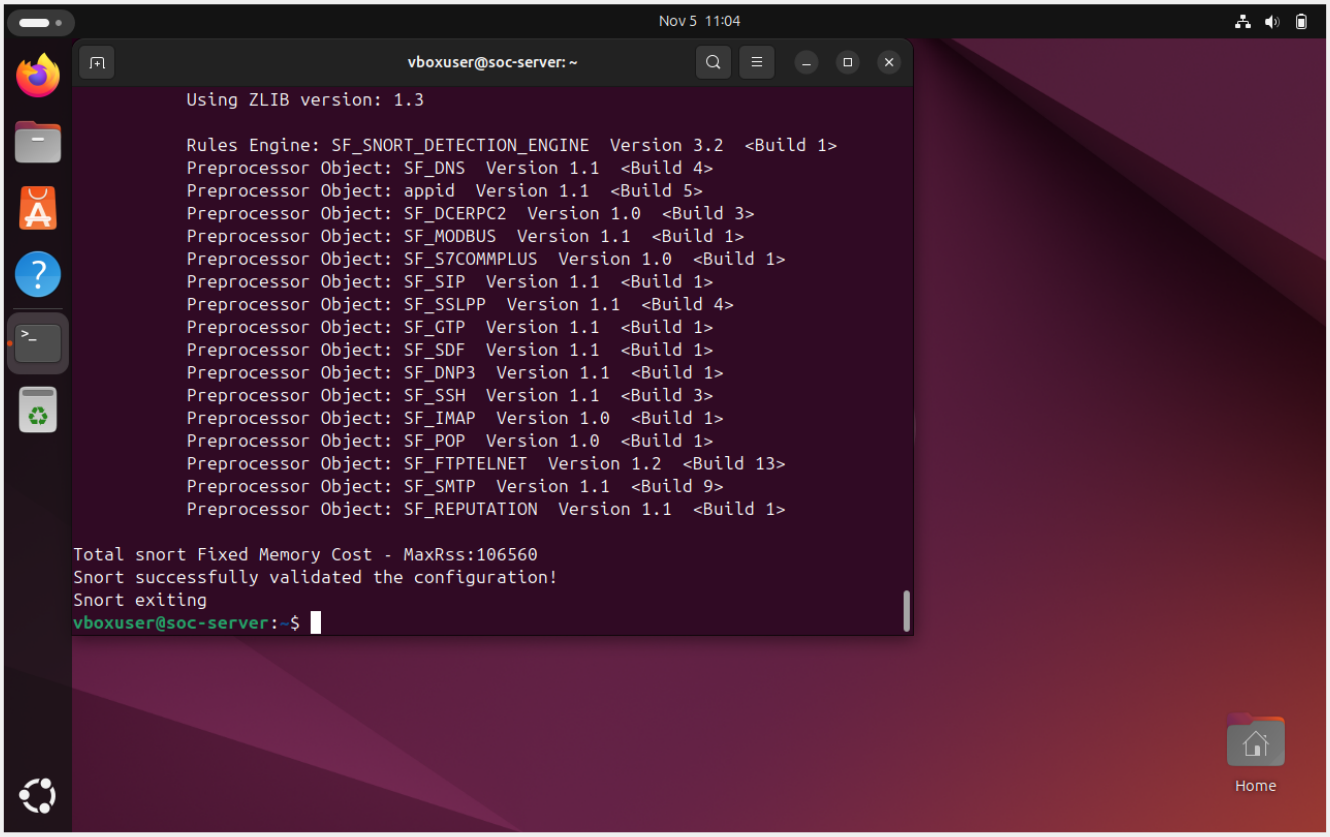


Figure 4. Successful Snort configuration validation output.

Validation Command

```
sudo snort -T -c /etc/snort/snort.conf -i enp0s3
```

The terminal explicitly reports: "Snort successfully validated the configuration!" This confirms the configuration parsed successfully at the time of the screenshot; it does not prove custom alerts were generated.

9. Custom Detection Rules

Task 4 | Rules prepared for implementation

Snort Rule Structure

A rule defines an action, protocol, source address/port, direction, destination address/port and options such as message, SID and revision.

ICMP (Ping) Detection

```
alert icmp any any -> any any (
 msg:"SOC ICMP Ping Detected"; sid:1000001; rev:1;)
```

This rule is designed to alert on ICMP packets visible to the monitored interface.

HTTP (Port 80) Detection

```
alert tcp any any -> any 80 (
 msg:"SOC HTTP Traffic Detected";
 flow:to_server,established; sid:1000002; rev:1;)
```

This detects matching established TCP traffic to port 80, not encrypted HTTPS traffic on port 443. The rules are proposed and have not yet been verified in the screenshots.

Implementation

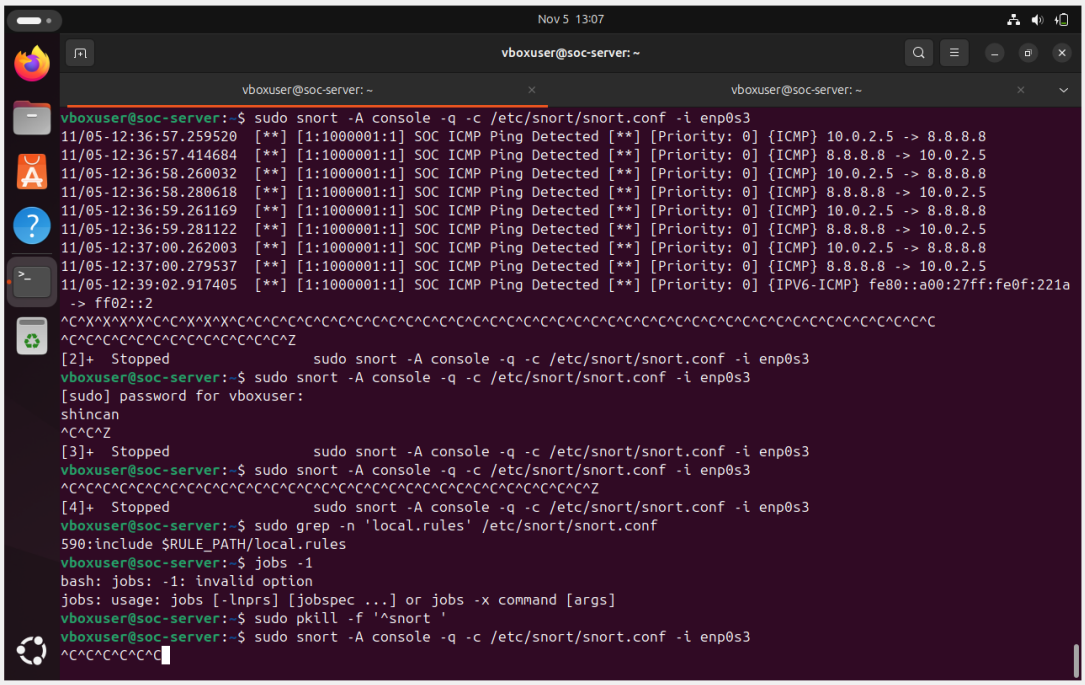
```
sudo nano /etc/snort/rules/local.rules
sudo grep "include.*local.rules" /etc/snort/snort.conf
```

10. Practical Detection Evidence

Live traffic monitoring on Ubuntu VM / Snort 2.9.20

VERIFIED 01 / ICMP PING DETECTION

Test: ping -c 4 8.8.8.8 | Interface: enp0s3



Observed Snort alert: SOC ICMP Ping Detected | SID 1000001

Traffic observed: 10.0.2.5 <-> 8.8.8.8 (ICMP request and reply).

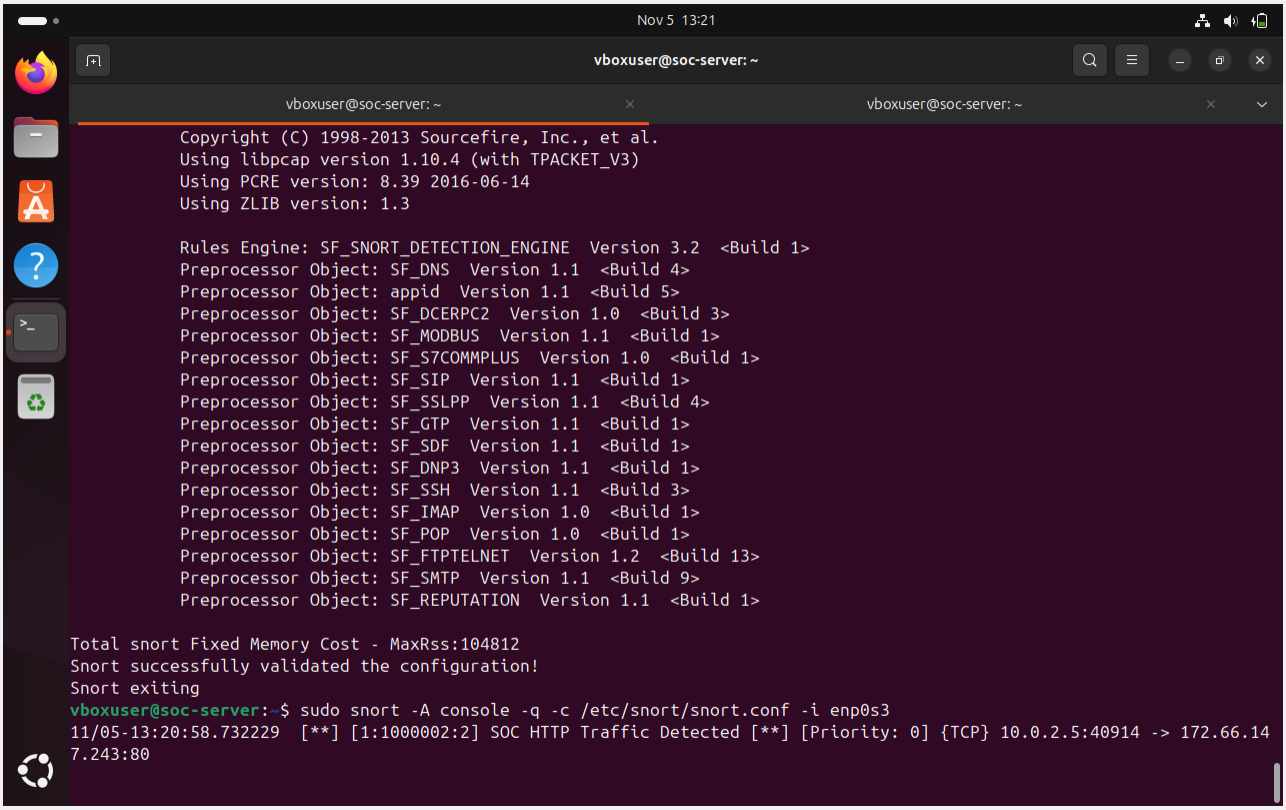
RESULT: PASS - live ICMP detection confirmed by supplied console evidence.

11. HTTP Traffic Detection

Evidence captured after generating HTTP traffic with curl

VERIFIED 02 / HTTP TRAFFIC DETECTION

HTTP request: curl -v --max-time 15 http://example.com/



Observed alert: SOC HTTP Traffic Detected | SID 1000002

Source: 10.0.2.5:40914 -> Destination: 172.66.147.243:80

Rule detects established TCP connections directed to port 80.

RESULT: PASS - Snort generated a live HTTP alert.

12. Validation & Test Results

Evidence-based outcomes and testing matrix

Ubuntu virtual lab VM screenshots supplied	PASS
Snort installation Version 2.9.20 shown	PASS
HOME_NET configuration 10.0.2.0/24 configured	PASS
Configuration validation Successful validation output	PASS
Local rules inclusion snort.conf includes local.rules	PASS
ICMP detection Live SID 1000001 alerts	PASS
HTTP detection Live SID 1000002 alert	PASS

Scope and limitations

Evidence confirms live alerts for ICMP and HTTP on the monitored interface.

This is a small IDS demonstration, not a full enterprise SOC deployment.

The screenshots show Ubuntu Desktop in VirtualBox; a separate Ubuntu Server installation is not independently evidenced.

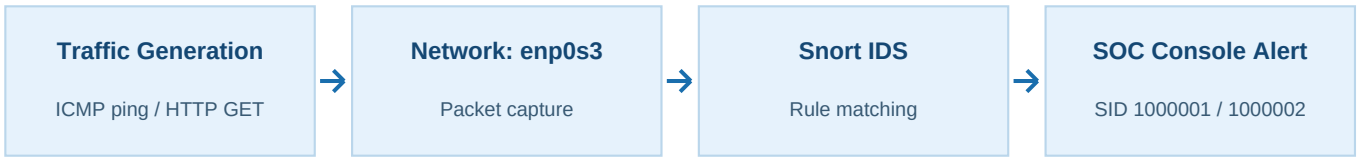
13. Conclusion & Submission

Completed practical work / final review

PROJECT OUTCOME

- Snort was installed and its configuration validated successfully.
- Two custom rules were configured in local.rules.
- ICMP ping traffic triggered the custom alert (SID 1000001).
- HTTP port 80 traffic triggered the custom alert (SID 1000002).
- Terminal screenshots document the practical detection results.

How the monitoring workflow operates



Traffic is captured, inspected against local rules, and reported as a console alert.

FINAL CONCLUSION

The lab demonstrates working signature-based intrusion detection.

Both assigned traffic categories were tested with real console alerts.

The assignment provides conceptual coverage and reproducible evidence.

Future work: centralized logging, dashboards, and alert triage.

Reference: Snort documentation - https://www.snort.org/documents

Reference: Ubuntu documentation - https://documentation.ubuntu.com/